

Gaurav Deshmukh

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SUMMARY

Software & AI engineer with 3+ years building production data pipelines, ML systems, and backend tooling, hardened by deep systems work on AI-accelerator silicon at Intel. Designs, trains, and deploys ML models against real-time data, and pairs full-stack software fluency (Python, C/C++, SQL) with the reliability and performance discipline of low-level hardware engineering.

SKILLS

Languages: Python, C/C++, SQL, Java, Bash

Backend & Infra: PostgreSQL, Kubernetes, Docker, REST APIs, ETL pipelines, Linux,, Grafana, Git, CI/CD, Azure

ML & AI: TensorFlow, Keras, Scikit-learn, model training & deployment, LLM tooling, MCP / AI agents

EXPERIENCE

Data & Systems Engineer

September 2025 – Present

FYM Technologies

- Architect end-to-end pipelines that ingest, normalize, and stream live telemetry from Toyota GT4, Ferrari GT3, and Porsche GT3 cars to engineers within seconds, from device integration to data store.
- Engineered the system for partial failures with retries, backpressure, and monitoring, holding 98% reliability across an entire race season under real-time, high-stakes conditions.
- Design, train, and deploy ML regression models against live telemetry streams, turning raw sensor data into real-time strategy and performance predictions for engineering teams.

FPGA Solutions Engineer

August 2022 – August 2025

Intel Corporation

Santa Clara, CA

- Built and owned C/C++ and Python software tooling across 9+ FPGA and NPU silicon programs, driving automated regression, system-level debug, and validation across firmware, host-side drivers, and their integration.
- Co-developed a new FPGA transactor that, working with the NPU, implemented a new inter-IP communication layer for next-generation AI accelerator silicon, increasing communication efficiency by 170%.
- Designed and maintained Kubernetes-hosted dashboards for build, debug, and performance tracking, giving 3+ engineering teams real-time visibility into system performance.
- Wrote Python/SQL automation for log ingestion and analysis, eliminating 10 hours of manual overhead per week.

Graduate Technical Intern

June 2021 – October 2021

Intel Corporation

Santa Clara, CA

- Architected a PostgreSQL telemetry pipeline extracting silicon log data and routing 100+ daily records for downstream analysis.
- Designed and deployed an interactive Grafana dashboard visualizing FPGA build, run, and debug processes, reducing manual analysis time by 160%.
- Implemented a Keras regression model forecasting system behavior to optimize hardware workflows.

PROJECTS

CarManac — Global Vehicle Database

April 2026 – Present

- Designing a global production-vehicle database on PostgreSQL + pgvector with per-fact provenance tracking.
- Modeling a five-level vehicle hierarchy from make down to variant, with an attribute table for long-tail specs.

AI-Driven Yellow Flag Predictor

January 2021 – Present

- Engineered a Keras neural network binary classifier to generate live yellow-flag predictions at 75% accuracy.
- Built a Selenium pipeline scraping real-time IMSA timing data into MongoDB; deployed the model for live race events.
- Integrating live timing-service APIs for richer real-time data access, collaborating with WRL officials to improve redmist.racing timing in the process.

EDUCATION

B.S. Computer Science (AI Specialization), Minor in Management • UC Irvine • 2022 • GPA: 3.6